

INF 354

Ionic I [LP07]

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How Ionic, Angular, and APIs Work Together

Ionic: The Face (UI)

Provides the mobile-friendly buttons, lists, and cards that the user sees.



User Interaction

The process begins when a user interacts with the Ionic UI.

Angular: The Brain (Logic)

Manages app structure, routing, and data communication via services.



Angular Request

Angular handles the logic and sends a request to the API.

API: The Data Source (Backend)

A C# backend that processes requests and retrieves information from the database.



API Response

The API sends JSON data back for Ionic to display nicely.

```
{  
  "data":  
    "value"  
}
```

```
{  
  "data":  
    "value"  
}
```

Technology Stack Quick Reference

Layer	Technology	Primary Role
Frontend UI	Ionic	Visual components (ion-button, ion-list)
Frontend Logic	Angular	Interaction handling and API calls
Backend	API (C#)	Data storage and request processing

Ionic I

Learning Outcome:

- Understand, develop, and deploy mobile applications using the Ionic framework and TypeScript.

Learning Objective:

- To understand the purpose of using Ionic.
- To develop an Ionic project.



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Outline

- Introduction
- Architecture
- Ionic Project Structure
- Setup
- CLI
- Ionic Components
- Ionic CSS Utilities
- Handling User Input
- Using Ionic Modals and Alerts
- Navigating Between Pages



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Introduction

- Ionic (7.2.1) is an open-source UI toolkit that includes over 100 components.
- It works with Angular, React, Vue, and other frameworks.
- It's cross-platform and has an adaptable user interface.
- The stunning open-source platform for creating hybrid mobile apps

```
Microsoft Windows [Version 10.0.22631.2428]
(c) Microsoft Corporation. All rights reserved.

C:\Users\u25738829>ionic -v
7.2.1
```



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Key Features of Ionic Framework

- Cross-Platform Development
- Integration with Angular
- Component Library
- Capacitor
- Performance
- Development Efficiency
- Community and Support



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Native vs Hybrid App

Native App

- Proficiency in each platform required
- Entirely separate code bases
- Timely & expensive development

Hybrid App

- HTML5 that acts like native
- Web wrapped in the native layer
- Direct access to native APIs



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Native vs Hybrid App

Native App	Hybrid App
Native apps are platform specific	Hybrid apps are platform independent
Respective development tools	It employs HTML5, CSS3, and JavaScript
Good performance	It has restricted performance
Time consuming apps	it can be produced more quickly than native apps..
Expensive app development	Using Capacitor, we can directly access native APIs

Examples of Native Apps:WhatsApp; Instagram; Uber

Examples of Hybrid Apps:Twitter; Facebook; Airbnb



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Prerequisite

To get started with Ionic Framework, the only requirement is

- Node

```
C:\Users\u25738829>node -v  
v22.13.1
```

Install from <https://nodejs.org/> [choose LTS – Long term support]

- npm environment [comes automatically installed with Node.js]

```
C:\Users\u25738829>npm -v  
11.2.0
```

- Visual Studio Code [The latest version of Visual Studio Code as of March 2025 is version 1.99]



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Ionic I - CL

To install the CLI: `npm install -g @ionic/cli`

- To create a project: `ionic start projectname template`
- To create an angular ionic app: `ionic start projectname tabs --type=angular`
- To test in a browser: `ionic serve`
- To see all templates available: `ionic start --list`



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Ionic Architecture

Ionic is based on standard web technologies:

- HTML (Structure)
- CSS (Sass/SCSS) (Styling)
- JavaScript/TypeScript (Logic)



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Ionic Core Building Blocks

UI Components

<ion-card>,<ion-img> etc

Themes & Styles

CSS & CSS Variables

Navigation

Custom or Built-in Router

State Management

Passing Data Around

Native Device Feature

user location ,camera
features and more.

Ionic CLI & publishing

Development and
Deployment



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Ionic Pros and Cons

Pros

- Single code base and Multi platform
- Development cost reasonable
- All components will be updated from web directly
- In app interaction has look and feel consistent
- Web Technology

Cons

- Run on web-View
- Base on plugins
- May require native related coding in device related issues.

Ionic Project Structure

Overview of Ionic project folders and files:

src/ - Main application source code

app/ - Contains core app components and modules

assets/ - Holds images and static files

pages/ - Contains individual application pages

theme/ - Custom styling and theming files



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CLI (Command Line Interface)

- Used to create, build, and run Ionic applications.
- Common Ionic CLI commands:
 - ionic start myApp - Creates a new Ionic project
 - ionic serve - Runs the app in a development server
 - ionic generate - Generates components, pages, and services
 - ionic build - Builds the application for deployment



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Ionic Components

- Pre-built UI components for mobile-friendly interfaces
- Commonly used components:
 - ion-button (Buttons)
 - ion-input (Text Inputs)
 - ion-list (Lists)
 - ion-card (Cards)
 - ion-grid (Grid Layout)



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Ionic CSS Utilities

- Predefined CSS classes to style applications
- Key features:
 - Text Styling (text-center, text-bold)
 - Padding & Margin (p-4, m-2)
 - Flexbox Utilities (ion-align-items-center, ion-justify-content-between)



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Handling User Input

- Forms and user interactions
- Ionic form controls:
 - ion-input for text fields
 - ion-checkbox for checkboxes
 - ion-radio-group for radio buttons
 - ion-toggle for switches
- Handling events with (ionChange), (ionInput), etc.



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Using Ionic Modals and Alerts

- Modals: Used for displaying pop-up dialogs
 - ion-modal for custom modals
 - ModalController to control modals
- Alerts: Used for notifications and confirmations
 - ion-alert component
 - Configurable buttons and inputs



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Navigating Between Pages

- Navigation using IonRouterOutlet and ion-nav
- Programmatic navigation with Angular Router
 - `this.router.navigate(['/page-name']);`
- Back navigation with NavController
 - `this.navCtrl.back();`



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Demo

1. Go to cmd prompt – type [ionic start]
2. Framework = Angular
3. Project name: ionic-weather
4. Select – blank: starter project
5. Select – NgModules
6. Type: cd ionic-weather
7. Navigate to [home.page.html](#) //follow the code
8. Navigate to [home.page.scss](#) //follow code



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Demo

- Open cmd screen/prompt:

```
(c) Microsoft Corporation. All rights reserved.

C:\Users\u25738829>ionic start
? Use the app creation wizard? No

Pick a framework!

Please select the JavaScript framework to use for your new app. To bypass this prompt next time, supply a value for the
--type option.

? Framework: Angular

Every great app needs a name!

Please enter the full name of your app. You can change this at any time. To bypass this prompt next time, supply name,
the first argument to ionic start.

? Project name: myApp

Let's pick the perfect starter template!

Starter templates are ready-to-go Ionic apps that come packed with everything you need to build your app. To bypass this
prompt next time, supply template, the second argument to ionic start.

? Starter template:
  tabs      | A starting project with a simple tabbed interface
  sidemenu  | A starting project with a side menu with navigation in the content area
> blank     | A blank starter project
  list      | A starting project with a list
  my-first-app | A template for the "Build Your First App" tutorial
```

Demo

Standalone components are the default way to build with Angular that simplifies the way you build your app.
To learn more, visit the Angular docs:
<https://angular.dev/guide/components>

NgModules

```
? .\myApp exists. Overwrite? Yes
✓ Preparing directory .\myApp in 6.11s
✓ Downloading and extracting blank starter in 995.06ms
> ionic integrations enable capacitor --quiet -- myApp io.ionic.starter
> npm.cmd i --save -E @capacitor/core@latest
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache i
f you want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerfu
l.
npm warn deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supported
npm warn deprecated glob@7.2.3: Glob versions prior to v9 are no longer supported

added 1133 packages, and audited 1134 packages in 2m

246 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
> npm.cmd i -D -E @capacitor/cli@latest

added 61 packages, and audited 1195 packages in 7s

254 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

Demo 1: Weather App

Learn more: <https://ion.link/enterprise>

```
> npm.cmd i
```

```
up to date, audited 1199 packages in 3s
```

```
254 packages are looking for funding  
  run `npm fund` for details
```

```
found 0 vulnerabilities
```

```
Join the Ionic Community! ❤️
```

```
Connect with millions of developers on the Ionic Forum and get access to live events, news updates, and more.
```

```
? Create free Ionic account? No
```

```
Your Ionic app is ready! Follow these next steps:
```

- Go to your new project: `cd .\myApp`
- Run `ionic serve` within the app directory to see your app in the browser
- Run `ionic capacitor add` to add a native iOS or Android project using Capacitor
- Generate your app icon and splash screens using `cordova-res --skip-config --copy`
- Explore the Ionic docs for components, tutorials, and more: <https://ion.link/docs>
- Building an enterprise app? Ionic has Enterprise Support and Features: <https://ion.link/enterprise-edition>

```
C:\Users\u25738829>
```


Blueprint of an Ionic Weather Interface

Roboto Font Family

The interface utilizes 'Roboto' sans-serif as the primary font for a clean, modern look.

Dynamic Sky Gradient

Use a linear gradient transition from deep teal (#298089) to light blue (#60D6FA).

\$size1
(Primary Temperature)
Relative applications: 12em

\$size3
(City Name)
Relative application: 3em

\$size5
& Description)
Relative size: 1.5em

Extreme Font Scaling

Font sizes strictly categorized from 1em up to a massive 12em for temperatures.

12em

3em

1.5em

IONIC WEATHER APP

24°C

City Name

Date
Description

THU

FRI

SAT

THU

FRI

SAT

Hero Weather Display

A centered main block displays City, Date, Description, and Temperature in a vertical stack.

High-Contrast Header:

A translucent header with white text (#fff) provides clear branding against the gradient background.

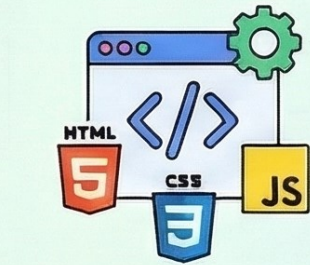
Fixed Bottom Forecast

An ion-grid is locked to the bottom of the screen to show secondary data.

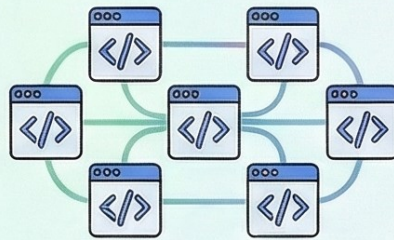
IONIC FRAMEWORK: BUILDING CROSS-PLATFORM HYBRID APPS

One Codebase. Multiple Platforms. Faster Development.

Hybrid App (Ionic)



Single codebase (HTML5/CSS3/JS) to run across multiple platforms independently.



Direct access via Capacitor

THE HYBRID ADVANTAGE



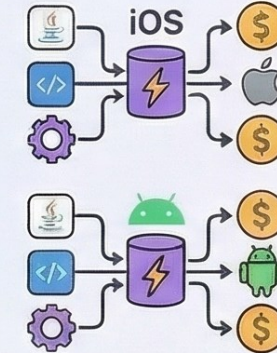
Faster & cost-effective

Hybrid Leaders



Proficiency in multiple platforms and separate, expensive codebases for each.

Native App Complexity

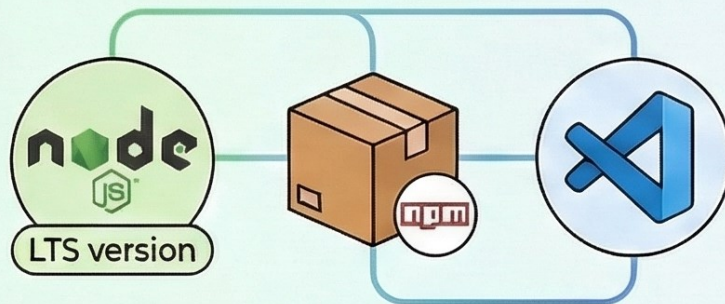


Timely & expensive

Native Examples

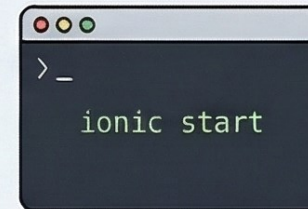
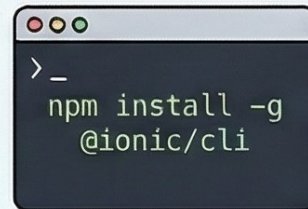


GETTING STARTED WITH IONIC



CORE PREREQUISITES

Development requires Node.js (LTS version), npm, and Visual Studio Code.



ESSENTIAL CLI COMMANDS

Install via npm and create projects using the Ionic CLI.

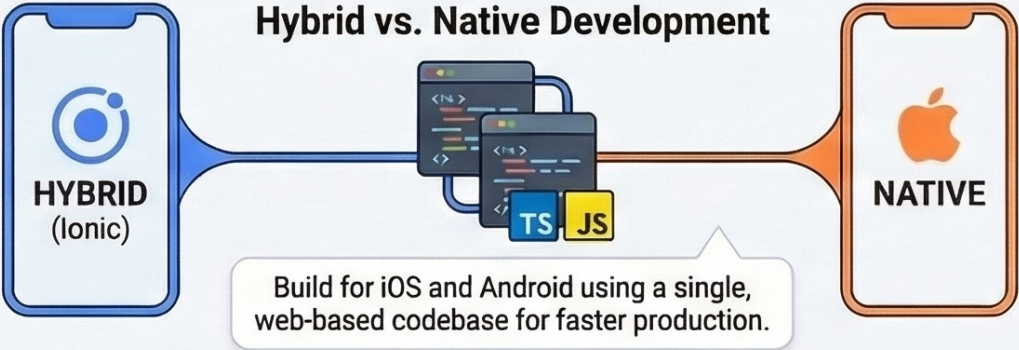


100+ PRE-BUILT COMPONENTS

Access a massive library of mobile-friendly UI elements like cards, buttons, and grids.

Ionic Framework: Mastering Hybrid Mobile Development

HYBRID DEVELOPMENT & ADVANTAGES



FEATURE	NATIVE APP	HYBRID APP (Ionic)
Codebase	Platform Specific	Single / Platform Independent
Development Speed	Time Consuming	Produced more quickly
Cost	Expensive	Reasonable / Cost-effective



STANDARD WEB TECH STACK

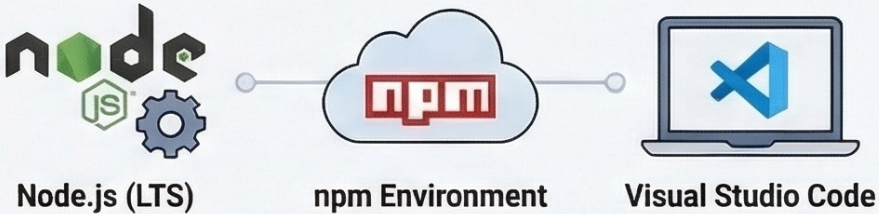
Utilizes standard web technologies including HTML, CSS (Sass), and TypeScript or JavaScript.



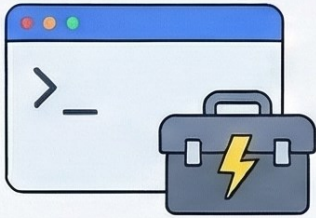
100+ PRE-BUILT UI COMPONENTS

Includes mobile-friendly building blocks like cards, buttons, grids, and input controls.

SETUP AND COMMAND ESSENTIALS



DEVELOPMENT PREREQUISITES



THE IONIC CLI TOOLKIT

A command-line interface used to create, build, and run mobile applications.

ESSENTIAL STARTER COMMANDS



Create projects



Browser testing

END



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